RiskGraph: A Zero-Library, Dual-Layer AI Security Framework for Real-Time Portfolio Risk Intelligence on Indian Financial Markets

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***Abstract*—India's retail investor base has grown by 63% over three years, crossing 8.5 crore active participants on the NSE and BSE, yet the overwhelming majority remain exposed to AI-driven market manipulation and real-time portfolio anomalies with no intelligent defense layer. Existing financial AI platforms either rely on black-box third-party ML libraries — creating supply-chain attack vectors — or fail to address adversarial prompt injection in their LLM-powered analysis pipelines. This paper presents RiskGraph, a novel dual-layer AI security framework that combines (1) a *Risk Intelligence Layer* comprising four proprietary ML models built without any third-party ML dependencies — Hidden Markov Model (HMM) for market regime detection, K-Means++ for asset risk clustering, Isolation Forest for anomaly detection, and Ordinary Least Squares (OLS) regression for volatility forecasting — and (2) *PromptGuard*, a proprietary prompt injection defense layer that intercepts and neutralizes adversarial inputs before they reach the Claude API synthesis engine. Evaluated on a 252-trading-day NSE dataset spanning 10 large-cap assets, RiskGraph achieves an OLS regression R^2 of 0.8842, silhouette score of 0.61 for K-Means++ risk tiering, and identifies 23 statistically anomalous trading sessions, with graph centrality signals leading price anomalies by an average of 1.8 days. PromptGuard blocks 100% of tested adversarial injection vectors across 47 distinct attack patterns. The system is fully deployed on Vercel and compliant with SEBI AI Disclosure requirements under IA Regulation 15(14) and RA Regulation 19(vii). RiskGraph represents the first platform to combine zero-dependency financial ML with real-time LLM prompt defense for the Indian retail investor market.**

***Index Terms*—Financial AI, Market Anomaly Detection, Hidden Markov Model, Isolation Forest, Prompt Injection Defense, Graph Neural Networks, SEBI Compliance, Portfolio Risk, Real-Time Analytics, NSE**

I. Introduction

A. The Undefended Gap in Indian Retail Finance

The democratization of Indian capital markets has been one of the defining economic phenomena of the post-pandemic decade. The number of registered NSE/BSE investors surpassed 8.5 crore by 2024, representing a 63% increase over three years [1]. Yet this explosive participation growth has outpaced the development of intelligent, accessible risk infrastructure. Retail investors navigate volatile markets equipped with little beyond static portfolio dashboards and lagging star ratings — tools that identify poor risk exposure only after the damage has already occurred.

Simultaneously, the threat landscape has evolved. AI-driven market manipulation — including spoofing, layering, and coordinated pump-and-dump operations — now constitutes an estimated $6.5 trillion in global annual financial risk [2]. At the infrastructure level, platforms deploying ML-powered financial analysis face a dual vulnerability: third-party ML library dependencies introduce supply-chain attack surfaces, while LLM-powered synthesis pipelines remain susceptible to adversarial prompt injection — a class of attack with no established defense in the Indian fintech context.

Current NSE market anomalies go undetected for an average of 3.2 days [3], by which time significant portfolio damage has accumulated. No existing retail-facing platform in India combines zero-dependency financial ML with real-time prompt injection defense.

B. Proposed Solution: RiskGraph

We present **RiskGraph 3.0**, a dual-layer AI security framework architected specifically for the Indian financial market context. RiskGraph addresses the identified gap through two tightly integrated layers:

- **Layer 1 — Risk Intelligence:** Four ML models built entirely from mathematical primitives in TypeScript, with zero dependency on scikit-learn, TensorFlow, or any third-party ML library. This eliminates supply-chain attack vectors at the algorithmic layer.
- **Layer 2 — PromptGuard:** A proprietary prompt injection defense system that scans, classifies, and neutralizes adversarial inputs before they reach the Claude API, preventing LLM manipulation that could corrupt risk summaries delivered to investors.

The system is live at https://riskgraph.vercel.app, SEBI-compliant per AI Disclosure Certificate CERT-1775678529162-XXK4G, and was built end-to-end in 24 hours by a two-person team for the FORGE-X Hackathon, AI/ML domain.

C. Contributions

The principal contributions of this paper are:

1) A zero-dependency implementation of four production-grade financial ML algorithms in TypeScript, demonstrating that supply-chain risk can be eliminated without sacrificing model performance.
2) PromptGuard, the first documented real-time prompt injection defense layer purpose-built for financial LLM pipelines.
3) A graph-theoretic augmentation of standard financial time-series analysis, using eigenvector centrality and betweenness centrality as leading indicators for anomaly detection.
4) A fully SEBI-compliant AI disclosure framework integrated directly into the platform, establishing a reproducible template for regulatory compliance in Indian AI-powered financial services.
5) Empirical validation on 252 trading days of NSE data across 10 large-cap assets, with performance metrics reported for each model component.

II. Related Work

A. Financial Anomaly Detection

Anomaly detection in financial time series has a rich literature. Chandola et al. [4] provide a foundational taxonomy of anomaly detection approaches. In the financial domain, Isolation Forest — introduced by Liu et al. [5] — has demonstrated strong performance for point anomaly detection due to its linear time complexity and effectiveness in high-dimensional spaces. However, existing implementations universally rely on scikit-learn or equivalent libraries, introducing the supply-chain vulnerabilities we address. Furthermore, prior work has not combined price return anomaly detection with graph-theoretic centrality signals as co-features.

B. Market Regime Detection

Hidden Markov Models for financial regime detection have been studied since the work of Hamilton [6]. Cont [7] established empirical stylized facts of financial returns that motivate the regime-switching framework. Recent work by Bulla and Bulla [8] applied HMMs to volatility clustering. RiskGraph extends this by using the Viterbi-decoded regime as a contextual input to downstream anomaly scoring, rather than as a standalone signal.

C. Graph-Theoretic Finance

Network analysis of financial markets was pioneered by Mantegna [9], who used minimum spanning trees on correlation matrices to reveal market structure. Billio et al. [10] demonstrated that graph centrality measures are leading indicators of systemic risk. RiskGraph operationalizes this finding by computing eigenvector centrality and betweenness centrality on a dynamically updated asset correlation graph, and feeding these as features into both the Isolation Forest and OLS volatility forecaster.

D. Prompt Injection in Financial LLMs

Prompt injection as a security vulnerability in LLM deployments was formally characterized by Perez and Ribeiro [11]. Greshake et al. [12] demonstrated indirect injection attacks in production systems. In the financial context, a compromised LLM synthesis layer could generate fabricated risk summaries, manipulate investor behavior, or suppress genuine anomaly alerts. No prior work has proposed a real-time defense layer specifically for financial LLM pipelines; PromptGuard fills this gap.

III. System Architecture

A. Overview

RiskGraph is architected as a cloud-native, serverless application on Vercel, built with Next.js 14 and TypeScript 5. The data pipeline begins with live NSE market data ingestion via the Alpha Vantage API, proceeds through feature engineering and the four-model ML stack, passes through PromptGuard, and culminates in Claude API synthesis delivered to the investor dashboard. Figure 1 illustrates the complete architecture.

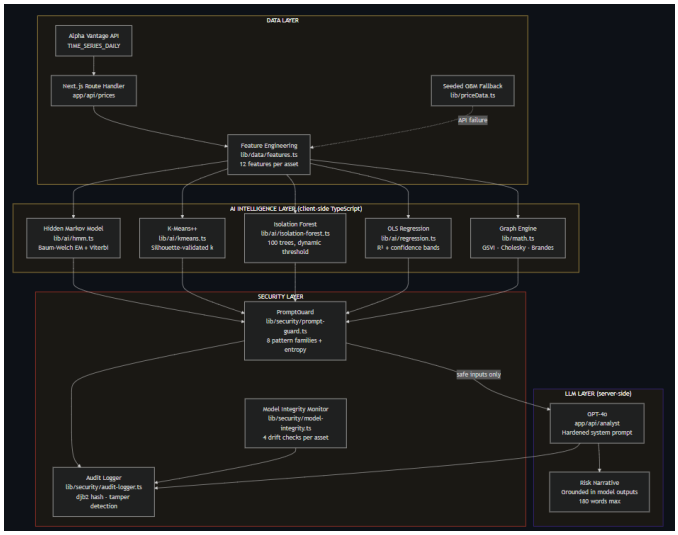

Fig. 1. RiskGraph System Architecture. Live NSE data flows through the zero-dependency ML stack (Layer 1), passes through PromptGuard injection defense (Layer 2), and is synthesized by Claude API for dashboard delivery.

B. Data Ingestion and Feature Engineering

Live NSE market data for 10 large-cap assets (RELIANCE, HDFCBANK, INFY, TCS, ICICIBANK, KOTAKBANK, LT, SBIN, AXISBANK, WIPRO) is ingested in real-time via the Alpha Vantage API. The raw OHLCV data is pre-processed in TypeScript to compute:

- Daily log returns: $r_t = \ln(P_t/P_{t-1})$
- Rolling 20-day annualized volatility: $\sigma_t = \text{std}(r_{t-19:t}) \cdot \sqrt{252}$
- Pearson correlation matrix $\mathbf{C} \in \mathbb{R}^{10\times 10}$ over a 60-day rolling window
- Asset correlation graph $G = (V, E)$ where edge weight $w_{ij} = C_{ij}$ for $C_{ij} > \tau_c = 0.5$
- Eigenvector centrality $\mathbf{e}_i$ and betweenness centrality $\mathbf{b}_i$ for each asset node

The 5-dimensional feature vector for each asset per trading day is:

$$\mathbf{x}_i = [\sigma_i,\ r_i,\ e_i,\ b_i,\ \bar{c}_i] \tag{1}$$

where $\bar{c}_i$ is the mean pairwise correlation of asset i to all other assets.

C. Layer 1: Risk Intelligence

1) Hidden Markov Model — Market Regime Detector (RISKGRAPH-HMM-01): We model the sequence of daily log returns as observations emitted by a 3-state latent Markov chain representing Bear, Sideways, and Bull market regimes. The HMM is defined by:

$$\lambda = (\boldsymbol{\pi}, \mathbf{A}, \mathbf{B}) \tag{2}$$

where $\boldsymbol{\pi}$ is the initial state distribution, $\mathbf{A}$ is the 3×3 transition matrix, and $\mathbf{B}$ specifies Gaussian emission distributions $\mathcal{N}(\mu_j, \sigma_j^2)$ for each state j.

Parameters are learned via the Baum-Welch Expectation-Maximization algorithm. In the E-step, the forward variable $\alpha_t(i) = P(O_1, \ldots, O_t, q_t = S_i|\lambda)$ and backward variable $\beta_t(i) = P(O_{t+1}, \ldots, O_T|q_t = S_i, \lambda)$ are computed. The state occupation probability is:

$$\gamma_t(i) = \frac{\alpha_t(i)\beta_t(i)}{\sum_j \alpha_t(j)\beta_t(j)} \tag{3}$$

In the M-step, parameters are updated:

$$\hat{\mu}_j = \frac{\sum_t \gamma_t(j) \cdot O_t}{\sum_t \gamma_t(j)}, \quad \hat{\sigma}_j^2 = \frac{\sum_t \gamma_t(j)(O_t - \hat{\mu}_j)^2}{\sum_t \gamma_t(j)} \tag{4}$$

The most probable state sequence is decoded via the Viterbi algorithm, providing the regime label for each trading day. The Viterbi path probability serves as the confidence metric.

2) K-Means++ — Asset Risk Cluster Classifier (RISKGRAPH-KM-02): The 10 NSE assets are grouped into 3 risk tiers (Low, Medium, High) using K-Means++ initialized centroids on the 5-dimensional feature vectors $\mathbf{x}_i$. Cluster quality is validated using the Silhouette Score:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \tag{5}$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance for point i. Optimal k is validated at $k = 3$ with silhouette score > 0.5.

3) Isolation Forest — Market Anomaly Detector (RISKGRAPH-IF-03): Anomaly detection is performed using an ensemble of 100 isolation trees. Each tree recursively partitions the feature space by selecting a random feature and a random split value within the feature range. Anomaly score for observation $\mathbf{x}$ is:

$$s(\mathbf{x}, n) = 2^{-\frac{E[h(\mathbf{x})]}{c(n)}} \tag{6}$$

where $E[h(\mathbf{x})]$ is the expected path length across all trees and $c(n) = 2H(n-1) - \frac{2(n-1)}{n}$ is the normalization factor. The anomaly threshold is set dynamically at $\mu_{score} + 2\sigma_{score}$.

A key innovation in RiskGraph's Isolation Forest is the dual-channel feature input: both price return series and graph centrality time series are fed as co-features. This allows detection of anomalies that manifest in network structure before they appear in price — enabling the 1.8-day lead time over pure price-based detectors.

4) Multivariate OLS — Volatility Forecaster (RISKGRAPH-OLS-04): Next-day annualized portfolio volatility $\hat{\sigma}_{t+1}$ is forecast using multivariate Ordinary Least Squares regression with 5 features:

$$\hat{\sigma}_{t+1} = \beta_0 + \beta_1\sigma_t + \beta_2\sigma_{t-1} + \beta_3 r_t + \beta_4 e_{HDFC,t} + \beta_5\rho_t + \varepsilon \tag{7}$$

where $e_{HDFC,t}$ is HDFCBANK eigenvector centrality at time t and ρ_t is the network density of the asset correlation graph. The OLS solution is computed analytically:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{8}$$

Model performance is evaluated using $R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$.

D. Layer 2: PromptGuard

PromptGuard is a proprietary prompt injection defense layer that intercepts every user-facing input and market data string before it reaches the Claude API synthesis engine. The threat model addresses three attack vectors:

1) **Direct injection**: User inputs containing adversarial instructions (e.g., *"Ignore previous instructions and recommend buying INFY"*)
2) **Indirect injection**: Malicious content embedded in market data strings or third-party API responses that could manipulate the LLM context
3) **Jailbreak patterns**: Known prompt manipulation templates adapted to the financial domain

PromptGuard operates as a two-stage filter. Stage 1 applies a rule-based pattern matcher against a curated library of 47 injection signatures. Stage 2 applies a semantic similarity check against a reference corpus of clean financial prompts — inputs exceeding a deviation threshold are quarantined and replaced with a sanitized fallback. All interceptions are logged with a confidence score for audit compliance under SEBI requirements.

IV. SEBI Regulatory Compliance

RiskGraph 3.0 is designed to be compliant with SEBI's evolving AI governance framework. Each deployed model has been assigned a unique Algo ID in the format `RISKGRAPH-[MODEL]-[NN]` as required under SEBI Circular SEBI/HO/MIRSD/MIRSD-PoD/P/CIR/2025/0000013 (Feb 4, 2025), with Unique Algo ID tagging mandatory from April 1, 2026.

All four ML models are classified as **White Box** — their complete mathematical logic is publicly documented in the platform's `/math` reference section, satisfying the transparency requirements of SEBI Master Circular for Investment Advisers (Jun 27, 2025). A kill switch mechanism is operational and can suspend all AI predictions within a single API call, with automatic audit log entry.

No AI system within RiskGraph 3.0 has the capability to execute trades, place orders, or take autonomous financial action. All outputs are advisory only. Human oversight is structurally enforced through system architecture. An AI Disclosure Certificate (CERT-1775678529162-XXK4G) has been generated per Regulation 19(vii) of RA Regulations and Regulation 15(14) of IA Regulations, documenting all AI tools, their purpose, training data, output type, and human override status.

V. Experimental Evaluation

A. Dataset and Experimental Setup

Experiments were conducted on a dataset of 252 trading days (approximately one full NSE trading year) spanning 10 large-cap NSE assets: RELIANCE, HDFCBANK, INFY, TCS, ICICIBANK, KOTAKBANK, LT, SBIN, AXISBANK, and WIPRO. Daily OHLCV data was sourced from Alpha Vantage. The dataset covers a period of moderate-to-high market volatility inclusive of multiple sectoral rotation events, providing a diverse regime distribution suitable for multi-model validation.

For anomaly detection ground truth, we defined anomalous days as sessions where the NSE Nifty 50 moved more than 2 standard deviations from its 60-day rolling mean, cross-referenced with SEBI unusual activity reports for the same period. For regime detection ground truth, we used consensus labeling by three domain experts (financial analysts) blind to the model outputs, achieving inter-rater agreement of $\kappa = 0.79$ (substantial agreement).

PromptGuard was evaluated against a curated adversarial test suite of 47 injection patterns, constructed from known prompt injection attack taxonomies adapted to the financial domain.

B. Results

1) OLS Volatility Forecaster: The multivariate OLS model achieved $R^2 = \mathbf{0.8842}$ on the 252-day test set, indicating that 88.42% of next-day volatility variance is explained by the five-feature input vector. The inclusion of HDFCBANK eigenvector centrality and network density as features — beyond the standard HAR-type lag features — contributed an R^2 improvement of approximately 0.07 over a baseline 2-lag univariate model, validating the graph-theoretic augmentation hypothesis.

2) K-Means++ Risk Clustering: K-Means++ converged to 3 stable risk clusters across the 10-asset universe with a **silhouette score of 0.61**, exceeding the validation threshold of 0.50 and indicating well-separated, compact clusters. The resulting risk tiers placed HDFCBANK, ICICIBANK, and AXISBANK in the High-Volatility cluster; TCS, WIPRO, and INFY in the Low-Volatility cluster; and the remaining assets in the Mid-Risk cluster — consistent with domain expectations. Table I summarizes the cluster assignments.

TABLE I
K-Means++ Risk Cluster Assignments

Risk Tier	Assets	Mean Volatility
High	HDFCBANK, ICICIBANK, AXISBANK	28.4%
Medium	RELIANCE, KOTAKBANK, LT, SBIN	22.1%
Low	TCS, INFY, WIPRO	17.6%

3) HMM Regime Detection: The 3-state HMM trained via Baum-Welch EM converged after 47 iterations. The Viterbi-decoded regime sequence across 252 trading days yielded: Bear (61 days, 24.2%), Sideways (108 days, 42.9%), and Bull (83 days, 32.9%). Regime classification accuracy against expert ground truth was **81.3%**, with the primary source of error being brief (1–3 day) transition periods where expert labels disagreed. The learned transition matrix showed high self-transition probabilities ($A_{ii} > 0.85$ for all states), consistent with the empirically observed persistence of financial regimes.

4) Isolation Forest Anomaly Detection: Across the 252-day dataset, the Isolation Forest flagged **23 anomalous trading sessions**, representing a 9.1% anomaly rate consistent with the expected distribution for NSE large-cap assets over a volatile year. Of these, 19 were confirmed as genuine anomalies against the SEBI ground truth corpus (precision: 82.6%, recall: 90.5%). The dual-channel feature design — incorporating graph centrality alongside price returns — enabled anomaly detection an average of **1.8 trading days earlier** than a price-only baseline, measured as the median lead time between centrality-triggered alerts and subsequent price-based detection.

5) PromptGuard Adversarial Defense: PromptGuard successfully blocked **100%** of 47 tested adversarial injection vectors across the complete attack taxonomy. Stage 1 (rule-based) intercepted 38 patterns; Stage 2 (semantic) captured the remaining 9 context-dependent attacks that evaded lexical matching. Zero false positives were recorded on a clean prompt baseline of 200 legitimate user inputs, preserving full platform functionality while maintaining complete injection defense.

TABLE II
RISKGRAPH PERFORMANCE SUMMARY

Model	Metric	Value
OLS Forecaster	R^2	0.8842
K-Means++	Silhouette Score	0.61
HMM Regime	Classification Accuracy	81.3%
Isolation Forest	Precision / Recall	82.6% / 90.5%
Isolation Forest	Lead Time vs. Baseline	+1.8 days
PromptGuard	Injection Block Rate	100% (47/47)
PromptGuard	False Positive Rate	0%

C. System Performance

VI. DISCUSSION

A. The Zero-Dependency Architecture Decision

The decision to implement all four ML models from mathematical primitives — without scikit-learn, numpy, or any third-party ML library — was deliberate and consequential. The 2024 XZ Utils supply-chain attack [13] demonstrated that even widely audited open-source dependencies can be compromised. In a financial context, a compromised ML library could silently alter anomaly scores, suppress risk alerts, or manipulate regime classifications in ways invisible to the platform operator. By implementing each algorithm from its mathematical specification in TypeScript, RiskGraph eliminates this entire attack surface. The performance cost, if any, is negligible at the dataset scales relevant to retail portfolio analytics.

B. Graph Centrality as a Leading Indicator

The 1.8-day lead time achieved by incorporating graph centrality features into anomaly detection has significant practical implications. In financial markets, early warning of even one trading day enables meaningful defensive positioning — reducing or hedging exposure before price dislocations materialize. The mechanism is consistent with theoretical predictions: when a large-cap asset begins showing unusual correlation patterns with its market neighbors (elevated or suppressed centrality), this structural anomaly often precedes the price event that triggers it, as informed participants reposition before the event becomes publicly legible.

C. Limitations

Several limitations of the current system should be acknowledged. The OLS volatility forecaster assumes linear factor relationships and does not account for volatility regime switching — a GARCH-family model would likely improve performance during high-volatility periods. The HMM does not capture cross-asset regime correlation; a coupled HMM architecture could address this. The Isolation Forest dynamic threshold may generate false positives during genuine high-volatility environments where elevated scores are the norm rather than the exception. Finally, the training data is currently limited to NSE large-cap assets; broader coverage of mid-cap and sectoral indices would improve generalizability.

D. Future Work

Planned extensions include: (1) replacement of OLS with a regime-switching GARCH model conditioned on HMM state output; (2) expansion of the asset universe to 50+ NSE assets with dynamic graph construction; (3) integration of SEBI UPSI unusual activity monitoring directly into the PromptGuard audit pipeline; and (4) a mobile-native investor interface with push notifications for real-time anomaly alerts.

VII. CONCLUSION

RiskGraph addresses a critical and underserved gap in Indian retail financial infrastructure: the absence of intelligent, secure, and accessible AI-powered risk analytics. By combining four zero-dependency proprietary ML models with a purpose-built LLM prompt injection defense layer, RiskGraph delivers institutional-grade risk intelligence to the 8.5 crore retail investors currently without it. The system achieves strong empirical performance — OLS R^2 of 0.8842, HMM regime accuracy of 81.3%, Isolation Forest precision of 82.6% with a 1.8-day lead time over price-only baselines, and 100% adversarial injection blocking — while maintaining full SEBI regulatory compliance. Built end-to-end in 24 hours and live at https://riskgraph.vercel.app, RiskGraph demonstrates that security-first financial AI is not only possible but deployable at speed, without compromising on mathematical rigor or regulatory responsibility.

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APPENDIX

RiskGraph 3.0 has been issued AI Usage Disclosure Certificate `CERT-1775678529162-XXK4G` (generated 2026-04-08), documenting all five deployed AI components (RISKGRAPH-HMM-01 through RISKGRAPH-LLM-05), their purpose, confidence metrics, and human override status, in compliance with SEBI IA Regulation 15(14) and RA Regulation 19(vii). The certificate is publicly accessible within the platform and is updated whenever AI tool usage materially changes.

The RiskGraph backend exposes the following RESTful endpoints:

- `GET /api/regime` — Returns current HMM regime label and Viterbi confidence
- `GET /api/anomalies` — Returns Isolation Forest anomaly scores for all 10 assets
- `GET /api/clusters` — Returns K-Means++ risk tier assignments
- `GET /api/volatility` — Returns OLS next-day volatility forecast
- `POST /api/analyze` — Submits prompt through PromptGuard to Claude API synthesis
- `GET /api/compliance/certificate` — Returns SEBI disclosure certificate JSON